

0/19 Questions Answered

hw7-diff-props-single-mean

Student Name

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Q1
0 Points

Sleep deficiency can affect your physical, cognitive, and social functions. Sleep deprivation has been linked to many chronic health problems including obesity and depression. [A study](#) was conducted in 2008 to investigate sleep deprivation among United States residents.

	California Washington	
Reported sleep deprivation for past 30 days	924	2188
Did not report sleep deprivation for past 30 days	10621	19908

Consider the above table which depicts the number of randomly sampled residents surveyed that did and did not report sleep deprivation for the past 30 days by state.

Let

- p_1 represent the proportion of California residents that reported sleep deprivation in this sample
- p_2 represent the proportion of Washington residents that reported sleep deprivation in this sample
- π_1 represent the corresponding population parameter for California
- π_2 represent the corresponding population parameter for Washington

Save Answer

Q2**5 Points**

We want to calculate a 90% CI for the difference in proportion of residents that experienced sleep deprivation for the past 30 days between California and Washington (i.e. $\pi_1 - \pi_2$).

Q2.1**1 Point**

What is the point estimate for the difference in proportions? Enter only your final answer. Round to four decimal places.

[Save Answer](#)**Q2.2****1 Point**

What is the critical value for the 90% confidence interval? Enter only your final answer. Round to four decimal places. Include a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q2.3**1 Point**

What is the standard error for the confidence interval? Include a picture of your work, including the formula you used.



 Please select file(s)

Select file(s)

Save Answer

Q2.4**1 Point**

With the point estimate, critical value, and standard error we estimate the 90% CI for the difference in proportion of residents experiencing sleep deprivation for past 30 days, between California and Washington, to be $(-0.0243, -0.0137)$.

We will change the question slightly. Now we want to calculate a 90% CI for the difference in proportion of residents experiencing sleep deprivation for past 30 days, between Washington and California (i.e. $\pi_2 - \pi_1$). Note the order of the groups has changed now.

Think about which parts of the confidence interval formula change when you change the order of the difference in proportions.

The lower bound of the new confidence interval is: (Enter only your final answer and round to four decimal places)

The upper bound of the new confidence interval is: (Enter only your final answer and round to four decimal places)

Save Answer

Q2.5**1 Point**

What do you notice about the relationship between the two confidence intervals we get when changing the order of the groups?

Save Answer

Q3**6 Points**

Now we want to conduct a hypothesis test to test if there is evidence of a difference in proportions, with a significance level of 0.1.

$$H_0 : \pi_1 = \pi_2$$

$$H_A : \pi_1 \neq \pi_2$$

Q3.1**1 Point**

Calculate the pooled proportion assuming the null hypotheses is true ($H_0 : \pi_1 = \pi_2$). Enter only your final answer and round to four decimal places. Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q3.2****1 Point**

Calculate the standard error for the pooled proportion. Include a picture of your work including the formula used.

 Please select file(s)**Select file(s)****Save Answer**

Q3.3**1 Point**

If the null hypothesis were true, what would be the sampling distribution of the difference in sample proportions?

[Save Answer](#)**Q3.4****1 Point**

Now we need to calculate the p-value so we can judge if there is a statistically significant difference in proportions. Enter only your final answer and round to four significant digits. (You can enter scientific notation such as 2.436×10^{-2} as 2.436e-2). Then upload a picture of your work.

 Please select file(s)[Select file\(s\)](#)[Save Answer](#)

Q3.5**2 Points**

You should have found a p-value that is less than 0.1. Select the following option that is an appropriate conclusion, at a significance level of 0.1, from the results of our hypothesis test.

We accept the null hypothesis and conclude there is no evidence of a difference in proportion of residents experiencing sleep deprivation for the past 30 days between California and Washington.

We fail to reject the null hypothesis and conclude there is no evidence of a difference in proportion of residents experiencing sleep deprivation for the past 30 days between California and Washington.

We fail to reject the null hypothesis and conclude there is evidence of a difference in proportion of residents experiencing sleep deprivation for the past 30 days between California and Washington.

We reject the null hypothesis and conclude there is evidence of a difference in proportion of residents experiencing sleep deprivation for the past 30 days between California and Washington.

We reject the null hypothesis and conclude there is no evidence of a difference in proportion of residents experiencing sleep deprivation for the past 30 days between California and Washington.

Save Answer

Q4**2 Points**

We forgot to check that the assumptions were met for the confidence interval and hypothesis test we performed! Please identify the conditions and explain if they were met.

[Save Answer](#)

Q5**8 Points**

Consider the following hypothetical situation:

A researcher wants to investigate the nightly average amount of sleep a California residents gets. A general recommendation is to get 7-9 hours of sleep a night. The researcher wants to test is there is evidence, at a significance level of 0.05, of the nightly average amount of sleep a California resident gets being different than 8 hours. She surveys 100 California residents and get a sample mean of 7.75 with a standard deviation of 0.25.

Q5.1**1 Point**

Consider the hypothesis testing procedure.

The null and alternative hypothesis are:

$$H_0 : \mu = 8$$

$$H_A : \mu \neq 8$$

What is the point estimate? Enter only your final answer.

[Save Answer](#)

Q5.2**1 Point**

What is the standard error of the point estimate? Enter only your final answer. Include a picture of your work.

 Please select file(s)**Select file(s)****Save Answer****Q5.3****1 Point**

If the null hypothesis were true, what is the sampling distribution?

Save Answer**Q5.4****2 Points**

Now we need to calculate the p-value so we can judge if there is evidence of the average sleep being different than 8. Enter only your final answer and round to four significant digits. Then upload a picture of your work.

 Please select file(s)**Select file(s)****Save Answer**

Q5.5**2 Points**

You should have calculated a p-value much smaller than the significance level, 0.05. Explain to your friend Lukas, who loves sleeping, what this p-value means, in context. Tell him if there is evidence, based on this data, that the average amount of sleep a California resident gets is different than 8 hours.

[Save Answer](#)**Q5.6****1 Point**

Say the researcher then takes another sample. With the new data of 100 residents the sample mean is still 7.75 hours, but the sample standard deviation this time is 2.5. State how the p-value would change and explain why this change makes sense.

[Save Answer](#)

Q6**0 Points**

- Have you done the homework by yourself, with a peer(s), or used tools like ChatGPT? If you do not acknowledge any help from a person or ChatGPT, we will consider this cheating.

- If you provided help to a peer(s) and/or got help from your peers(s) please explain briefly what this entailed.

- If you used ChatGPT, include the prompts you used and explain how ChatGPT supported your learning.

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